

User and Group Management in Linux

useradd command :

1. `sudo useradd username` -> Creates a new user with default settings (no password set, no home directory unless configured by default on the system).
2. `sudo useradd -m username` -> Creates a new user along with a home directory, e.g. /home/username.
3. `sudo useradd -d /path username` -> Creates a user with a custom home directory instead of the default location.
4. `sudo useradd -s /bin/bash username` -> Sets the login shell for the user, for example bash or zsh.
5. `sudo useradd -c "Full Name" username` -> Adds a comment field, usually used for the user's full name.
6. `sudo useradd -u 1050 username` -> Creates a user with a specific UID instead of the next available one.
7. `sudo useradd -g groupname username` -> Sets the user's primary group at the time of creation.
8. `sudo useradd -G group1,group2 username` -> Adds the user to one or more secondary groups at creation.
9. `sudo useradd -e YYYY-MM-DD username` -> Sets an expiry date after which the account is automatically disabled.
10. `sudo useradd -r username` -> Creates a system account, normally used for services rather than real people.
11. `sudo useradd -p password username` -> Sets an encrypted password directly during user creation.

In Simple Words :

1. useradd username -> basic user creation
2. -m -> makes home directory
3. -d -> custom home directory
4. -s -> sets login shell
5. -c -> full name / comment
6. -u -> custom user ID
7. -g -> primary group
8. -G -> secondary groups
9. -e -> account expiry date
10. -r -> system account
11. -p -> set password

usermod command :

1. `sudo usermod -aG groupname username` -> Adds the user to an additional group without removing them from existing ones.
2. `sudo usermod -l newname oldname` -> Renames an existing user account.
3. `sudo usermod -s /bin/zsh username` -> Changes the login shell of an existing user.
4. `sudo usermod -d /newpath -m username` -> Moves the user's home directory to a new location.
5. `sudo usermod -L username` -> Locks a user's password so they cannot log in.
6. `sudo usermod -U username` -> Unlocks a previously locked user account.
7. `sudo usermod -e YYYY-MM-DD username` -> Changes the expiry date of an existing account.

In Simple Words :

1. -aG -> add to group
2. -l -> rename user
3. -s -> change shell
4. -d -m -> move home directory
5. -L -> lock account
6. -U -> unlock account
7. -e -> change expiry date

userdel command :

1. `sudo userdel username` -> Deletes a user account but keeps their home directory.
2. `sudo userdel -r username` -> Deletes a user account along with their home directory and mail spool.
3. `sudo userdel -f username` -> Forcefully deletes a user account even if they are currently logged in.

In Simple Words :

1. userdel username -> remove user, keep home folder
2. -r -> remove user and home folder
3. -f -> force remove

groupadd command :

1. `sudo groupadd groupname` -> Creates a new group with default settings.
2. `sudo groupadd -g 1050 groupname` -> Creates a group with a specific GID instead of the next available one.
3. `sudo groupadd -r groupname` -> Creates a system group, normally used for services rather than real people.

In Simple Words :

1. groupadd groupname -> basic group creation
2. -g -> custom group ID
3. -r -> system group

groupmod command :

1. `sudo groupmod -n newname oldname` -> Renames an existing group.
2. `sudo groupmod -g 2050 groupname` -> Changes the GID of an existing group.

In Simple Words :

1. -n -> rename group
2. -g -> change group ID

groupdel command :

1. `sudo groupdel groupname` -> Deletes an existing group, as long as it is not the primary group of any user.

In Simple Words :

1. groupdel groupname -> remove a group